

Python Course for Physicists (And everyone else!)

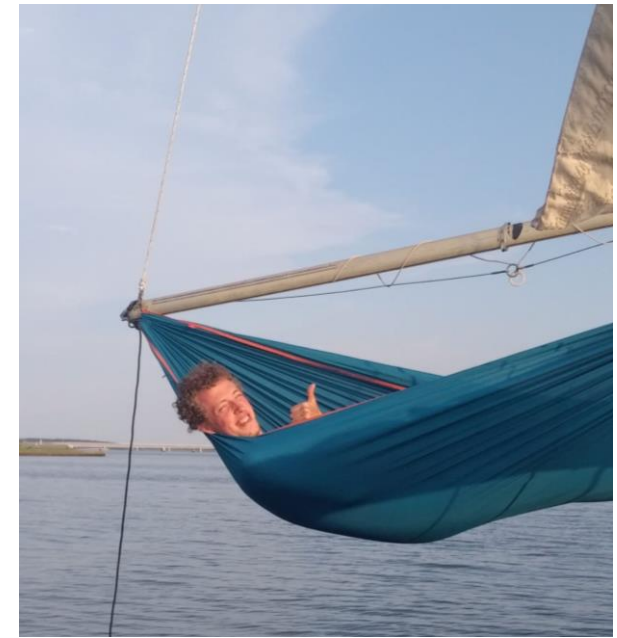
Chunk 1: Introduction & Setup

By Jonathan Bollig

08.10.2025

Background & About Me

- jonathan.bollig@uni-jena.de
- Master: Physics & Photonics
- Bachelor: Physics in Jena
- Self-taught Python with YouTube before studying
 - Super useful in physics
- Tried to convince all my friends
- Now a course for everyone else
- First time, let's see how it goes 😊



Hearing
my friends
complain about
SciDavis,
Excel and Origin

Helping
my friends
after switching
to python

The Tutors

- **Erik Prenzel** (erik.prenzel@uni-jena.de)
 - Student in Physics Master
 - Friend of Jonathan
 - Learned Python during my studies, 100% worth it
- **Hinrich Dirksen** (hinrich.dirksen@uni-jena.de)
 - Currently pursuing my experimental physics master thesis
 - I use Python everyday from data analysis+vis. to small simulations and automations in the lab
 - 3x better code since GitHub copilot arrived



The Tutors

- **Philipp Janzen** (philipp.janzen@uni-jena.de)
 - Physics Master's student
 - Friend of Jonathan
 - Learned Python during my studies
- **Verena Menhofer** (verena.menhofer@uni-jena.de)
 - Physics Master's student
 - Another friend of Jonathan
 - Started learning Python with YouTube, but mostly learnt as needed during studies



About You: (Experimental) Questionare



<https://vibes.fh-muenster.de/apps/live-votings/XETR2/>

General Information – Times and Dates

Dates:

- **Regular: Wednesday to Friday**
- Maybe Saturday if delayed or for extra “chunks” (Improvised if demanded)

Times:

- **Start at 09:30**
- Lunchbreak 12:30 – 13:30
- Finish: ~16:30

Lecturing type:

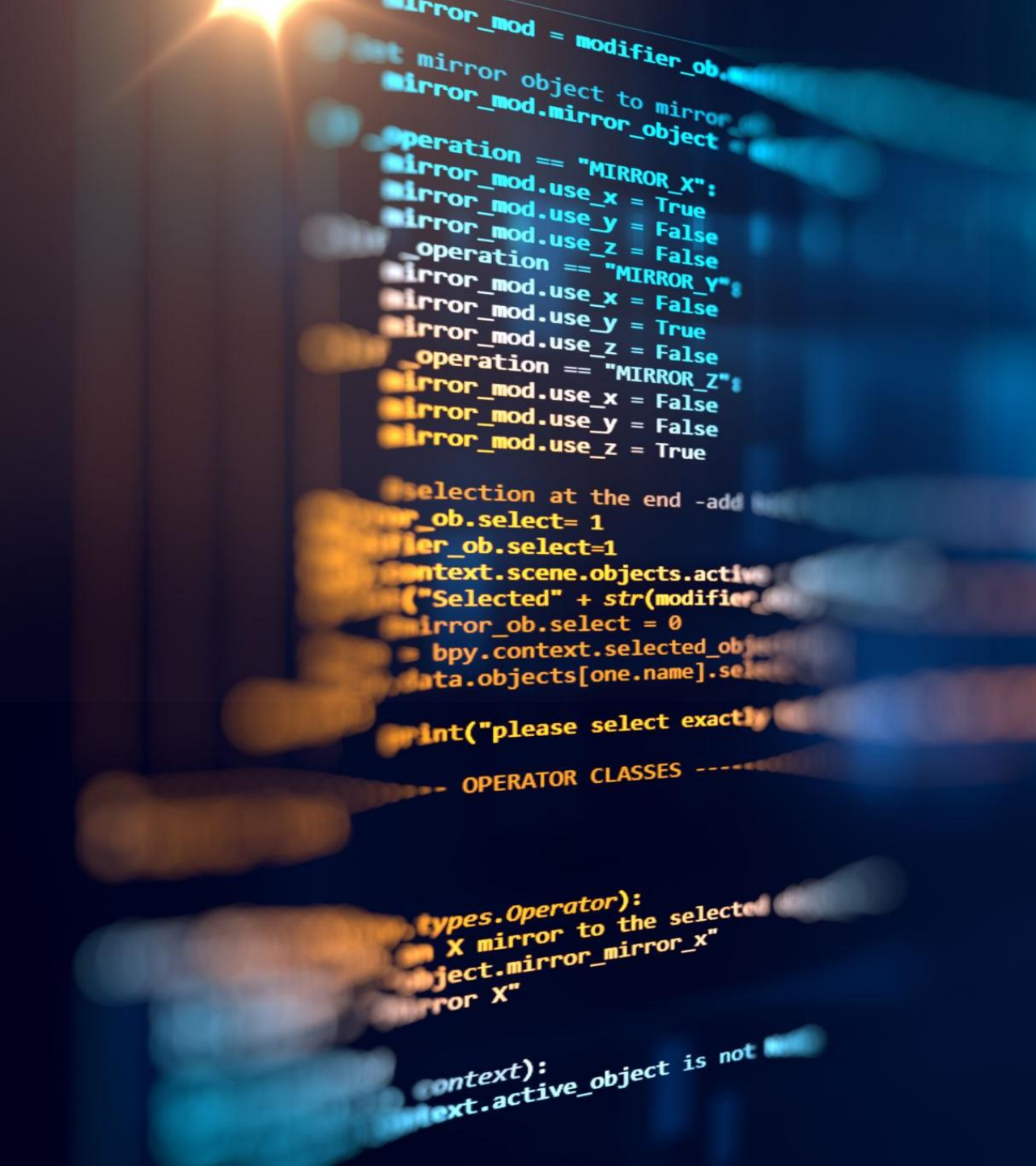
- 15-minute lectures (Chunks) with 45-minute practice sessions
- 3 chunks before and 3 after lunch → 18 chunks in total
- Exception: Now

General Information – Lecturing Type

- **15 Minute Lectures:** Concise introductions to topics
 - YouTube tutorial style
 - PDFs are uploaded
- **45 Minute self practice:**
 - Usage of Jupyter Notebooks: Tasks and solutions in one place.
 - Easy, medium and advanced tasks
- **Generally:**
 - Come and Go whenever you like
 - Ask Questions directly (Maybe everyone is wondering the same)
 - **ALL** Slides etc. are shared 😊

Why Coding?

- Data Analysis
- Visualization
- Simulations
- Automation
- Machine Learning
- ...

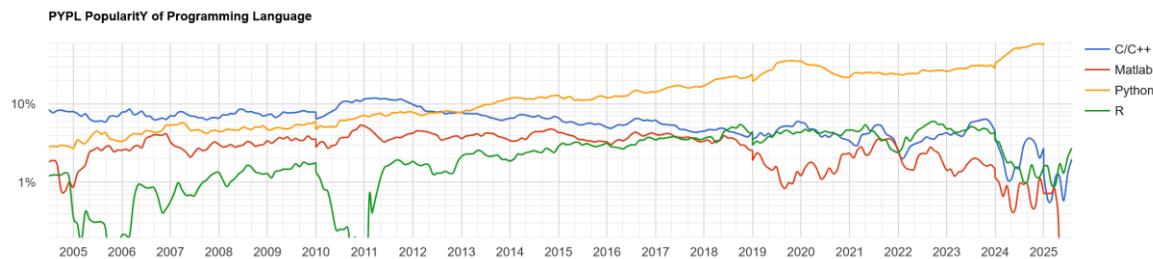


Own Examples

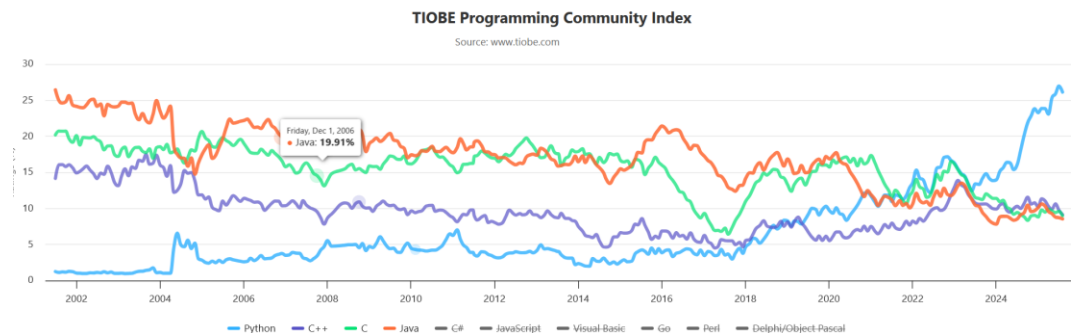
- Automated Experiment (UV-Experiment)
- Data Analysis from Lab (Oszi data from Experimental Optics)
- Irgendeine Simulation (vielleicht von PCQM)

Why Python?

- Most popular programming language world-wide
 - Web Development, AI / LLM, Data Science
- Popular in the scientific community (anecdotal evidence)
- Open Source / Free
- Easy to learn



[<https://pypl.github.io/PYPL.html>]



[<https://www.tiobe.com/tiobe-index/>]

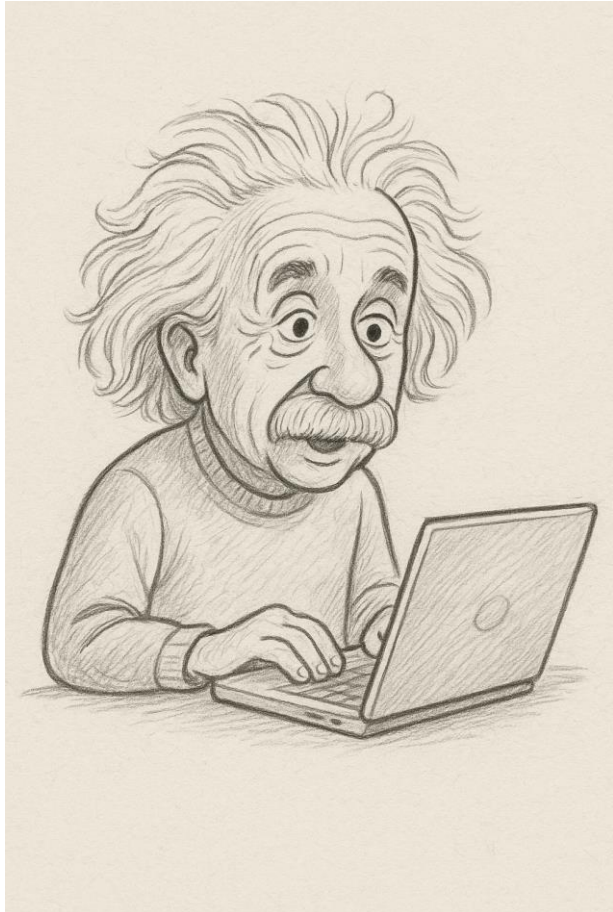
Comparison to other languages, (dis)advantages of Python

- Easy to learn
- Readable
- Slow but can be fast
- Popular

Learning goals

- Ability to use python as a tool in everyday (physics) life
 - Read and write data
 - Process and clean data
 - Visualize data
- We won't cover
 - Game Design
 - Webdesign
- What I might cover in the future:
 - Recursion
 - Version Control w. Git
 - Pretty Coding / PEP / Typing
 - Object Oriented Programming
 - Multithreading / Processing

What is a program?



[ChatGPT ☺]



```
1 print("Hello World")
2 print(2+2)
3 name = "Jonathan"
4 print("Hello " + name)
```

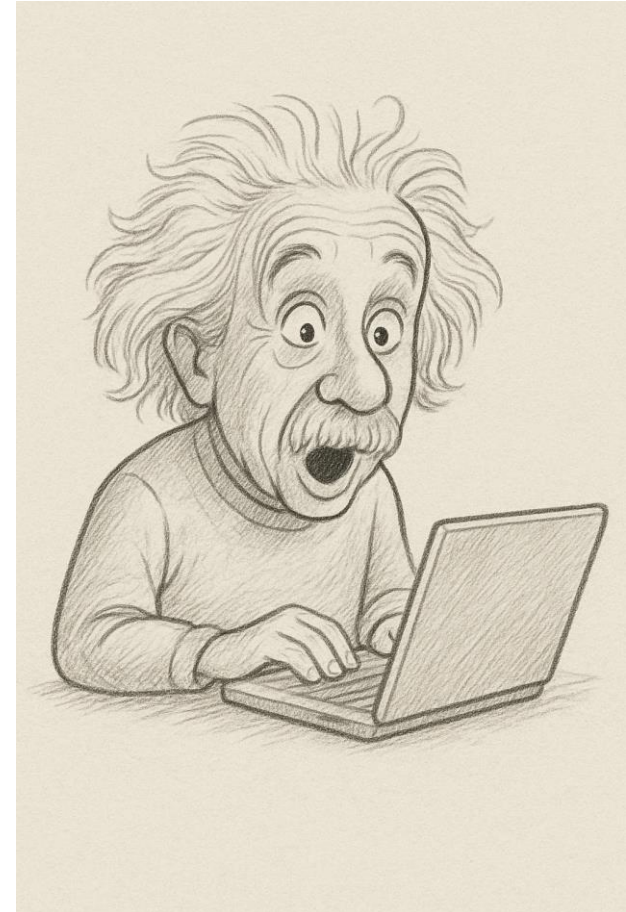


Some Magic ?

(Beyond our scope, google if interested)



```
Hello World
4
Hello Jonathan
```



[ChatGPT ☺]

2,5 Ways of Coding in Python

- Skript:

- Starts Python
- Runs Code from first to last line
- Finishes (closes Python) when last line is executed
- After execution: Memory lost

```
1 print("Hello World")
2 print(2+2)
3 name = "Jonathan"
4 print("Hello " + name)
```



```
Hello World
4
Hello Jonathan
```

- Interactive:

- Python is started
- Executes commands “on the go”
- Python is manually closed
- → Memory maintained

```
(base) PS C:\Users\jona> python
Python 3.12.7 | packaged by Anaconda,
Type "help", "copyright", "credits" or
>>> print("Hello World")
Hello World
>>> print(2+2)
4
>>> name = "Jonathan"
>>> print("Hello " + name)
Hello Jonathan
>>> quit()
(base) PS C:\Users\jona> |
```


2,5 Ways of Coding in Python: Notebooks

- **Jupyter Notebooks are interactive, but group code in cells:**
- **Advantages:**
 - Tasks, Cheat Sheets and place for solving all in one place.
 - No extra IDE needed.
- **Disadvantages:**
 - Sometimes stuff works, that shouldn't
 - Can promote bad coding practises
- In this course: Python + Notebooks treaded as one. At end of course: Introduction to other IDEs

Welcome to the example notebook

The first task is to print "Hello World", then print the sum of 2+2. After that, save your name in a variable, and print "Hello" and your name, using the variable.

[2]:

```
print("Hello World")
print(2+2)
name = "Jonathan"
print("Hello " + name)
```

```
Hello World
4
Hello Jonathan
```

The second task has not been thought of yet. Therefore do whatever you want.

[3]:

```
print("This is a very exciting comment")
```

```
This is a very exciting comment
```

Notebook Live Demo: Cells

- Markdown Cells

- Code Cells

- Cells can be:

- Moved
- Copy Pasted
- Etc.

Example Notebook

This is an example notebook. Execute it, to check if everything works.

This is a markdown cell

```
[4]: # This is a code cell. It can execute code:
print("Hello everyone :)")
print(2+2)
```

Hello everyone :)

4

A Big Header

A smaller header

A even smaller header

Some text

Some more text

A list of stuff:

- first stuff
- second stuff
- third stuff

```
[5]: a = 10
print(a)
```

10

Notebook Live Demo: Markdown Cells

```
# A Big Header  
## A smaller header  
### A even smaller header
```

Some text

Some more text

A list of stuff:

- first stuff
- second stuff
- third stuff



A Big Header

A smaller header

A even smaller header

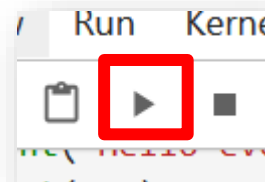
Some text

Some more text

A list of stuff:

- first stuff
- second stuff
- third stuff

- Execute Cell: Shift + Enter or



Notebook Live Demo: Code Cells

```
[1]: # This is a code cell. It can execute code:  
print("Hello everyone :)")  
print(2+2)
```



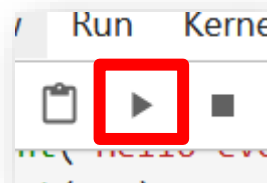
```
[2]: # This is a code cell. It can execute code:  
print("Hello everyone :)")  
print(2+2)  
  
Hello everyone :)  
4
```

- Number indicates execution
- Errors appear in Red
 - Pay attention to last line

```
[19]: impossible_task = "My age is " + 99
```

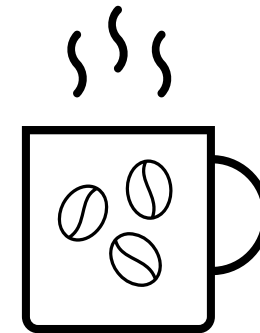
```
-----  
TypeError                                Traceback (most recent call  
Cell In[19], line 1  
----> 1 impossible_task = "My age is " + 99  
  
TypeError: can only concatenate str (not "int") to str
```

- Execute Cell: Shift + Enter or



Now you:

- Try to open the file “Tasks_1_2_example_notebook.ipynb” and execute the cells
 - In case of problems: Ask us
- Time: 30 Minutes (inkl. Coffee Break)



“The best way to learn a language is to speak to natives.”
The guy learning python:

